CHAPTER 16

STRUCTURE AND PHYSIOGRAPHY

CHAPTER ESSENCE AT A GLANCE:
Understand the three geological foundations of India (Peninsular Block, Himalayas, Indo-Ganga-Brahmaputra Plain), the parallel Himalayan longitudinal ranges, alluvium zones (Bhabar, Tarai, Bhangar, Khadar), and Peninsular plateaus.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Three Geological Provinces:** The rigid, stable Peninsular Block composed of ancient Archaean gneisses and granites; the young, flexible, and folded Himalayan mountain chain; and the Indo-Ganga-Brahmaputra geosynclinal alluvial trough.

• **Longitudinal Himalayan Architecture:** Three parallel ranges: Himadri (Greater Himalayas, continuous crystalline crest, average elevation 6000 m), Himachal (Lesser Himalayas, rugged valleys), and Shiwaliks (Outer Himalayas, uncompacted sediments with longitudinal flat-bottomed 'duns').

• **Northern Plain Sedimentological Stratification:** Sub-Himalayan depositional belts: Bhabar (porous pebble belt along foothill slopes where streams sink underground), Tarai (marshy, swampy re-emergence zone), Bhangar (older river terraces with calcareous kankar), and Khadar (young flood alluvium).

• **Peninsular Plateau and Coastal Realms:** Composed of Central Highlands and Deccan Plateau bounded by Western Ghats (continuous, steep scarp, high passes like Palakkad Gap) and Eastern Ghats (dissected by east-flowing rivers); flanked by Western and Eastern Coastal Plains.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

Sub-Himalayan Alluvial Succession: Bhabar (Pebbles) → Tarai (Marsh) → Bhangar → Khadar
Depositional Succession Across Northern Plain Foothills

Himalayan Longitudinal Elevation: Himadri (6000 m) ⟷ Himachal (4000 m) ⟷ Shiwalik (1000 m)
Hypsometric Longitudinal Hierarchy of the Himalayan Orogen

Peninsular Escarpment: Western Ghats (1500 m, Continuous) ⟷ Eastern Ghats (600 m, Broken)
Structural Asymmetry of Peninsular Flanking Escarpments

Island Genesis: Andaman and Nicobar (Submerged Ridges) ⟷ Lakshadweep (Coral Atolls)
Divergent Tectonic and Biogenic Genesis of Indian Islands

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Western Ghats are continuous and can be crossed only through specific natural passes (Thal Ghat, Bhor Ghat, Palakkad Gap), unlike the discontinuous, highly dissected Eastern Ghats cut by major east-flowing rivers.

• **Alert:** Khadar is young, highly fertile flood alluvium deposited every year by monsoon floods, whereas Bhangar is older alluvium that forms terraces above flood limits and contains calcium carbonate nodules (kankar).

• **Alert:** Streams disappear underground in the coarse, highly porous boulder-pebble gravel tract of the Bhabar and re-emerge without distinct channels in the Tarai to create a wet, swampy marshland.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between Bhangar and Khadar alluvium in the Northern Plains. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Bhangar:** Older alluvium deposited on higher terraces away from river beds, contains calcareous calcium carbonate concretions (kankar), and is less fertile. **[1 Mark]**

• **Khadar:** Newer, fine-grained alluvium deposited on the lower floodplains, enriched annually by seasonal monsoon flood silt, and is intensively fertile. **[1 Mark]**

2. What are 'Duns' in the Himalayan relief? Give two examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Flat-bottomed, longitudinal structural valleys formed between the Lesser Himalayas (Himachal) and the Outer Himalayas (Shiwalik ranges). **[1 Mark]**

• **Examples:** Dehra Dun, Kotli Dun, and Patli Dun. **[1 Mark]**

3. Compare the Western Ghats and the Eastern Ghats on the basis of continuity, elevation, and drainage. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Continuity:** Western Ghats form a continuous escarpment crossed only through passes like Thal Ghat and Palakkad Gap; Eastern Ghats are discontinuous and fractured. **[1 Mark]**

• **Elevation:** Western Ghats are higher (average height 900 to 1600 m, peaking at Anamudi 2695 m); Eastern Ghats are lower (average height 600 m, peaking at Mahendragiri). **[1 Mark]**

• **Drainage:** Western Ghats act as a primary water divide from which major rivers originate and flow eastwards; Eastern Ghats are dissected and breached by rivers like Godavari, Krishna, and Kaveri. **[1 Mark]**

4. Explain the characteristics of the Bhabar and Tarai tracts located along the foothills of the Himalayas. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Bhabar Tract:** Narrow belt (8 to 10 km wide) running parallel to Shiwalik foothills composed of coarse unassorted pebbles and boulders brought down by mountain streams; porosity is so high that streams sink underground. **[1.5 Marks]**

• **Tarai Tract:** Located south of Bhabar (10 to 20 km wide); underground streams re-emerge at the surface creating a damp, wet, swampy, and heavily forested marshy tract rich in wildlife. **[1.5 Marks]**

5. Describe the three parallel longitudinal ranges of the Himalayas, highlighting their elevation and physical characteristics. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Greater Himalayas (Himadri):** Northernmost, highest, and most continuous range; average elevation 6000 m; composed of Archaean granites and gneisses; features world's highest peaks like Mount Everest and Kanchenjunga. **[1.5 Marks]**

• **2. Lesser Himalayas (Himachal):** Rugged mountain system south of Himadri; elevation 3700 to 4500 m; includes famous ranges like Pir Panjal and Dhauladhar, and picturesque hill stations like Shimla and Darjeeling. **[1.5 Marks]**

• **3. Outer Himalayas (Shiwaliks):** Outermost low ranges; elevation 900 to 1100 m; composed of unconsolidated Tertiary sediments; separated from Himachal by structural flat-bottomed valleys called duns. **[1 Mark]**

6. Discuss the geological origin, structural subdivisions, and economic significance of the Peninsular Plateau of India. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Origin and Stability:** Formed as an ancient rigid part of Gondwanaland composed of crystalline granites and metamorphic gneisses; has remained stable without major marine submergence since the Cambrian. **[1 Mark]**

• **Structural Subdivisions:** (1) The Deccan Plateau (black soil volcanic basalt trap), (2) Central Highlands (Malwa and Bundelkhand plateaus), and (3) North-Eastern Extension (Meghalaya and Karbi Anglong plateaus). **[1.5 Marks]**

• **Economic Significance:** Known as the mineral heartland of India, containing vast reserves of iron ore, coal, manganese, bauxite, and mica in the Chotanagpur plateau; ideal for black soil cotton cultivation and hydroelectric generation. **[1.5 Marks]**